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047. Breast Cancer Classification and Segmentation Using Deep Learning on Ultrasound Images

Area of Research: Computer Vision, Medical Image Analysis, Oncology

Problem Statement: Breast cancer is one of the leading causes of death among women worldwide. Early detection plays a crucial role in reducing mortality rates and improving patient outcomes. Ultrasound imaging is a safe and widely used technique for breast cancer examination and early detection. However, the interpretation of ultrasound images can be challenging, even for experienced radiologists, due to the complexity and variability of breast lesions. The goal of this project is to develop a deep learning-based system for accurate classification and segmentation of breast masses in ultrasound images. By leveraging state-of-the-art deep learning techniques and a comprehensive dataset of breast ultrasound images, this project aims to assist radiologists in the early detection and diagnosis of breast cancer, potentially improving the efficiency and accuracy of the screening process.

Dataset:

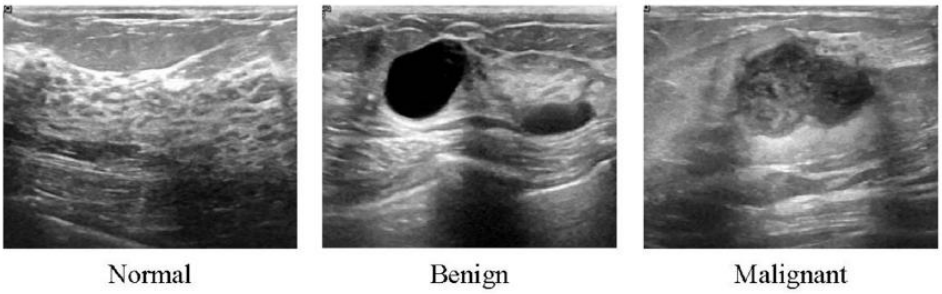

Figure 47: Samples of ultrasound breast images from the dataset, categorized into normal, benign, and malignant cases.

The Breast Ultrasound Dataset consists of 780 ultrasound images of breast masses, categorized into three classes: normal, benign, and malignant. The dataset was collected from 600 female patients aged between 25 and 75 years old at Baheya Hospital for Early Detection & Treatment of Women's Cancer in Cairo, Egypt. The images have an average size of 500 ? 500 pixels and are in PNG format. In addition to the original ultrasound images, the dataset also includes corresponding ground truth mask images for each lesion, enabling the development and evaluation of segmentation models.

Dataset URL: https://huggingface.co/datasets/gymprathap/Breast-Cancer-Ultrasound-Images-Dataset

Task: Develop a deep learning-based pipeline for breast cancer classification and segmentation using the Breast Ultrasound Dataset. Preprocess the ultrasound images, apply data augmentation techniques, and split the dataset into training, validation, and testing subsets. For the classification task, explore state-of-the-art deep learning architectures such as ResNet, DenseNet, or EfficientNet, and train a model to classify the images into normal, benign, and malignant categories. For the segmentation task, utilize semantic segmentation architectures like U-Net, DeepLab, or FCN, and train a model to accurately delineate the boundaries of breast masses in the ultrasound images.

Relevant Papers:

[1]. Al-Dhabyani, W., Gomaa, M., Khaled, H., & Fahmy, A. (2020). Dataset of breast ultrasound images. Data in Brief, 28, 104863. https://www.sciencedirect.com/science/article/pii/S2352340919312181

[2]. Yap, M. H., et al. (2018). Automated breast ultrasound lesions detection using convolutional neural networks. IEEE Journal of Biomedical and Health Informatics, 22(4), 1218-1226. https://ieeexplore.ieee.org/document/8003418

[3]. Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional networks for biomedical image segmentation. In International Conference on Medical Image Computing and Computer-Assisted Intervention (pp. 234-241). Springer, Cham. https://link.springer.com/chapter/10.1007/978-3-319-24574-4_28